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2nd Feb 2026 | DA6401

→ **Backpropagaⁿ in MLPs**

- SL → Data → $D = \{(x_i, y_i)\}_{i=1}^{n}$ → features, target

Model → funcⁿ that maps input x to y.
↳ fixed HPTs (no. of layers, no. of neurons)

Parameters → wts. & b, have to be learned.

Obj. / Loss funcⁿ → $J(w)$, diff. btw. $\hat{y}$ & y.

Learning algo → optimizaⁿ technique to reduce loss funcⁿ.

- Regression → predict conti. quantity.

Classificaⁿ → predict class probabilities (1 of K classes)
↳ 2 → binary classificaⁿ
↳ discriminative models $P(Y|X)$

Regress.
- Linear activaⁿ → $a(z) = z$ → pre-activaⁿ
- Loss funcⁿs
  - MSE → $L = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$ — penalize larger errors heavily
  - MAE → $L = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i|$ — robust to outliers

- Classificaⁿs → this one sum up to 1.

which of their output, prob. add up to 1?
1. Multi Class → mutually exclusive (1-hot)
   ↳ activaⁿ → softmax $\sigma(z) = e^{x_i} / \sum_j e^{x_j}$
2. Multi label → independent / non-exclusive
   ↳ sigmoid $\sigma(z) = \frac{1}{1+e^{-z}}$ → sigmoid for every output

- these activaⁿs works as the output layer

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- Multi Class

Loss → Categorical cross-entropy

total no. of classes

$$\boxed{L = -\sum_{c=1}^{C} y_c \log(\hat{y}_c)}$$

$y_c = 1$ for true class

Multi label

loss → binary cross-entropy (BCE)

total no. of classes

$$\boxed{L = -\frac{1}{M} \sum_{j=1}^{M} [y_j \log(\hat{y}_j) + (1 - y_j) \log(1 - \hat{y}_j)]}$$

→ summed over all output nodes

BCE → penalizes larger errors heavily, wrong predic^ns on both sides.